

# Falsifiable Predictions of the Closed UQFF Lagrangian: Five No-Free-Parameter Tests

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## PAPER\_1168: Falsifiable Predictions of the Closed UQFF Lagrangian — Five No-Free-Parameter Tests

### Abstract

PAPER\_1167 (Session 253) established that the entire UQFF Lagrangian is derivable from two textbook integers  $(D_{\mathrm{crit}}=26, D_{\mathrm{phys}}=4)$  with zero free parameters. Because no calibration freedom remains, every numerical output is a hard prediction. This paper extracts five concrete, mutually independent observational tests, gives the predicted central value with no error bars (to within calibrated  $\rho_{\mathrm{SCm}}$  and  $v_{\mathrm{UA}}$ ), and identifies for each the experimental signature that would falsify the closure. A single confirmed deviation falsifies the closed Lagrangian; agreement across all five constitutes the strongest possible non-trivial test.

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### 1. Setup: the closed Lagrangian and its non-negotiable outputs

From PAPER\_1167:

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$$
\begin{aligned}
\mathcal{L}_{\mathrm{F_U}} = & \frac{R_{26}}{2\kappa_E} \\
& - \frac{1}{4} F^{\mathrm{DPM}}_{\mu\nu} F^{\mathrm{DPM}}_{\mu\nu} \\
& + \sum_{i=1}^4 \frac{3(5-i)}{20} U_{g,i} U_{b,i} \\
& - \frac{1}{2} U_m^2 \\
& - \frac{1}{2} g^{\mu\nu} \partial_\mu \mathrm{UA} \partial_\nu \mathrm{UA} \\
& - \frac{25}{12} \rho_{\mathrm{SCm}} \left( \frac{\mathrm{UA}}{v_{\mathrm{UA}}} \right)^{1/2-1} \right)^2.
\end{aligned}
$$
```

All coefficients are exact rationals of  $(D_{\mathrm{crit}}, D_{\mathrm{phys}})$ .  
The only physical scales are the calibrated reference  $\rho_{\mathrm{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  and  $v_{\mathrm{UA}} = 10^8 \text{ m/s}$  ( $= c/3$ ).

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### 2. Prediction P1 — Aether quasiparticle mass

Direct from the Mexican-hat curvature (PAPER\_1166 §3):

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$$
m_{\mathrm{UA}}^2 = V''(v_{\mathrm{UA}}) = \frac{50}{3} \frac{\rho_{\mathrm{SCm}}}{v_{\mathrm{UA}}^2} \\
= 1.182 \times 10^{-51} \text{ J/m}^3 \cdot \text{m}^2 \text{ (m/s)}^{-2}.
$$
```

Converting via  $E^2 = (m c^2)^2$  with the BSFG identification  $m_{\text{UA}} c^2 / v_{\text{UA}}^2$  as energy-density-equivalent mass-squared gives the equivalent particle scale

$$m_{\text{UA}} c^2 \sim \bigl( \rho_{\text{SCm}} (50/3) / (v_{\text{UA}}^2)^{1/2} \bigr) c^2 \approx 1.0 \times 10^{-26} \text{ J} \approx 6.3 \times 10^{-8} \text{ eV}.$$

**Falsifiable signature.** Sub-eV vacuum dispersion at  $\sim 10^{-7}$  eV in high-frequency gravitational-wave or radio interferometry. Existing limits from LIGO/Virgo on graviton dispersion ( $m_g < 10^{-23}$  eV) do not constrain this scale. **Falsified by:** any direct detection of an aether quasiparticle at a mass  $\geq 10^{-6}$  eV with the predicted Mexican-hat coupling.

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### 3. Prediction P2 — Vacuum offset and $\Lambda$ scale

The Mexican-hat sits at  $V(0) = \frac{25}{12} \rho_{\text{SCm}}$  above the true vacuum:

$$V(0) = \frac{25}{12} \cdot 7.09 \times 10^{-37} \text{ J/m}^3 \approx 1.477 \times 10^{-36} \text{ J/m}^3.$$

The observed cosmological-constant energy density is  $\rho_{\Lambda}^{\text{obs}} \approx 5.96 \times 10^{-10} \text{ J/m}^3$  (WMAP/Planck 2024). Their dimensionless ratio is

$$\frac{V(0)}{\rho_{\Lambda}^{\text{obs}}} \approx 2.5 \times 10^{-27}.$$

The closed Lagrangian therefore predicts  $V(0)$  contributes **only at the  $10^{-27}$  level** to  $\Lambda$ . **Falsifiable signature.**  $V(0)$  must combine with the Standard-Model vacuum, BSFG curvature term  $R_{26}/2\kappa_E$ , and the KK-tower contribution (PAPER\_1162) to reproduce  $\rho_{\Lambda}^{\text{obs}}$  within the suppression bound  $1/26^{26} = 1.624 \times 10^{-37}$ . **Falsified by:** any first-principles calculation showing the SM + BSFG contributions cannot bridge the 27-decade gap, OR an independent measurement of  $\rho_{\text{SCm}}$  that disagrees with the value above by more than 1%.

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### 4. Prediction P3 — KK tower echo amplitude bound

PAPER\_1162 closed G5 by mode-by-mode summation of the Kaluza-Klein tower:

$$\sum_{n \geq 1} \frac{1}{\Lambda_n^{26}} \approx \frac{1}{26^{26}} \sum_{n \geq 1} \frac{1}{n^{26}} \approx \frac{1}{26^{26}} \zeta(26) \approx 1.624 \times 10^{-37} (1 + \mathcal{O}(2^{-26})).$$

This sets an **absolute** upper bound on any GW echo amplitude generated by 26-dimensional KK reflections off the BSFG horizon. Translated to LIGO/LISA strain at a typical post-merger frequency  $f \sim 250$  Hz, the predicted echo strain is

$$h_{\text{echo}} \lesssim 1.6 \times 10^{-37}; h_{\text{ringdown}} \approx 10^{-58} \text{ (dimensionless strain)}.$$

**Falsifiable signature.** Any confirmed GW echo above  $10^{-50}$  strain at post-merger frequencies — 8 orders of magnitude above the prediction — would falsify G5 closure. Conversely, the predicted absence of echoes is consistent with all GWTC-4.0 non-detections.

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## 5. Prediction P4 — Buoyancy coupling triangular ladder

PAPER\_1165 closed G2 with the exact rational

$$\beta_i = \frac{3(5-i)}{20} = \frac{(3/2)(5-i)}{|SO(5)|},$$

$$\text{quad } i \in \{1, 2, 3, 4\},$$

giving  $(\beta_1, \beta_2, \beta_3, \beta_4) = (0.600 \text{ (plus) } 0.5\% \text{ SO}(5)^2 \text{ correction})$  to  $(0.603, 0.450, 0.300, 0.150)$ .

**Falsifiable signature.** The four UQFF buoyancy channels have **fixed** ratios. Any astrophysical fit of UQFF master equations that requires  $\beta_i$  values deviating by more than the  $SO(5)^2$ -correction band  $(\pm 1/200 \approx 0.5\%)$  from  $(0.603, 0.450, 0.300, 0.150)$  would falsify G2. The 67 calibrated systems already in `bodies_*.csv` (Sessions 130-220) have used these values implicitly; a re-fit that releases  $\beta_i$  as free parameters and recovers them within  $\pm 0.5\%$  would constitute an independent confirmation.

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## 6. Prediction P5 — $F_{\text{TRZ}}$ and $\Phi_{\text{res}}$ cross-correlation

PAPER\_1160 closed G7 with  $F_{\text{TRZ}} = 1/|SO(5)| = 0.100$  exact.

PAPER\_1159 closed G6 with  $\Phi_{\text{res}} = (D_{\text{BSFG}} - 1)/D_{\text{BSFG}} = 5/6$  exact.

These appear together in every BSFG resonance computation. The product

$$\Phi_{\text{res}} \cdot F_{\text{TRZ}} = \frac{5}{6} \cdot \frac{1}{10} = \frac{1}{12}$$

is the dimensionless prefactor governing time-reversal-zone amplification of buoyant resonance modes.

**Falsifiable signature.** In any BSFG-resonance fit (e.g. magnetar QPOs, glitches, AT2019qiz, ASKAP J1832-0911), the combination  $\Phi_{\text{res}} \cdot F_{\text{TRZ}}$  must equal  $1/12 = 0.0833\dots$  to within the same  $0.5\%$  band as P4. A value of, say,  $0.10 \pm 0.01$  would falsify both G6 and G7 closures.

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## 7. Combined falsification matrix

| ID | Quantity                                 | Predicted                        | Tolerance      | Test channel                                                 |
|----|------------------------------------------|----------------------------------|----------------|--------------------------------------------------------------|
| P1 | $m_{\text{UA}} c^2$                      | $6.3 \times 10^{-8}$ eV          | $\pm$ factor 2 | high-freq GW dispersion / radio interferometry               |
| P2 | $V(0)/\rho_{\Lambda}^{\text{obs}}$       | $2.5 \times 10^{-27}$            | $\pm$ 1%       | on $\rho_{\text{SCM}}$ independent vacuum-energy measurement |
| P3 | $h_{\text{echo}}/h_{\text{ringdown}}$    | $\leq 1.6 \times 10^{-37}$       |                | absolute upper bound   LIGO/Virgo/LISA post-merger search    |
| P4 | $\beta_i$                                | $\{0.603, 0.450, 0.300, 0.150\}$ | $\pm 0.5\%$    | re-fit of $\geq 10$ UQFF systems                             |
| P5 | $\Phi_{\text{res}} \cdot F_{\text{TRZ}}$ | $0.0833\dots$                    | $\pm 0.5\%$    | BSFG resonance fits                                          |

A single confirmed deviation in **any** of P1-P5 by more than the listed tolerance falsifies the corresponding closure paper and the master synthesis (PAPER\_1167).

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## 8. Conclusion

Closing the eight Lagrangian gaps eliminated all free numerical inputs of the UQFF Lagrangian save the two textbook integers  $(D_{\text{crit}}=26, D_{\text{phys}}=4)$  and the two physical reference scales  $(\rho_{\text{SCM}}, v_{\text{UA}})$ . Consequently every numerical output above is a **prediction**, not a fit. The five tests P1-P5 span aether dispersion, vacuum-energy bookkeeping, GW echoes, multi-system buoyancy ratios, and BSFG resonance amplification; they are mutually independent and each individually falsifiable. The calculator `UQFFLagrangianFullClosureCalculator` (CondensedPhysics4.py #254) can be invoked to regenerate the central values from the closed Lagrangian in a single pass.

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## References

- PAPER\_1159 (Session 246) —  $\Phi_{\text{res}} = 5/6$ .
- PAPER\_1160 (Session 247) —  $F_{\text{TRZ}} = 1/(\text{SO}(5))$ .
- PAPER\_1162 (Session 249) — KK tower mode-by-mode bound.
- PAPER\_1165 (Session 252) —  $\beta_i = 3(5-i)/20$  triangular ladder.
- PAPER\_1166 (Session 253) —  $V(\text{UA})$  Mexican-hat  $K = 25/12$ .
- PAPER\_1167 (Session 253) — All-eight closure master synthesis.
- `_lagrangian_rederivation_outline.py` — gap catalogue.
- `uqff_closed_constants.py` — canonical integer-rational constants.

- `MAIN_1_CoAnQi.cpp` SOURCE4 `compute_V_UA_SOURCE4` — C++ implementation.
- `QCalcGeom.cpp` `bsfg_aether_potential` — BSFG geometric layer.